

Vikram Pande

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Education

North Carolina State University

Aug 2022 – May 2024

Master of Science in Electrical Engineering; GPA: 3.97/4.00

Raleigh, NC

Courses: Random Processes, Image Processing, Neural Networks, Advanced Machine Learning, Object Oriented Development, Pattern Recognition, Cloud Computing, Natural Language Processing, Computer Vision, Probabilistic Graphical Models

Savitribai Phule Pune University

Aug 2016 – May 2020

Bachelor of Engineering Electronics and Telecommunication; GPA: 7.93/10.00

Pune, India

Courses: Machine Learning, Image Processing, Data Structures & Algorithms, Computer Networks, Linear Algebra, Vector Calculus

Experience

Dentsply Sirona

June 2024 – Present

Machine Learning Engineer

Charlotte, NC

- **Led development and productionization** of ML systems and infrastructure, owning end-to-end codebases; **adopted by multiple** internal sales teams.
- Engineered and managed an **agentic coding assistant** using **Codex** and **Claude** with several **MCP servers** and **skills** that autonomously writes code, commits, pushes, and raises PRs, **reducing manual workflow steps** for data engineers by **~50%**.
- Built an **LLM-RAG-based agent** with a vector database for **semantic search** across 500+ product documents, handling 1,500+ monthly product queries with 85% accuracy
- Improved recommendation precision by **25%**, achieving **0.83 Precision@K** and **0.64 NDCG**, by designing a **two-tower neural recommender** system in **PyTorch**.
- Boosted churn prediction accuracy by **18%** on **100K+ records** using probabilistic models (BG/NBD) and RFM features; improved product sales forecasting accuracy by **20%** using statistical and ML models (ARIMA/Prophet) in **PySpark**.
- Reduced model deployment time by **40%** by implementing scalable **MLOps pipelines** for legacy and new models.

Sozzani Lab, NCSU

Sep 2023 – May 2024

Research Assistant

Raleigh, NC

- Achieved **96% F1-score** in **protein sequence classification** by training a custom **CNN-Attention-LSTM** model on Arabidopsis data; published in [Nature](#).
- Reduced model training time by **60%** by parallelizing **neural network training** using **High Performance Computing**.
- Developed an **Autoencoder & KMeans pipeline** to identify and cluster **plant subtypes** for improved phenotype mapping.

Syngenta

June 2023 – Dec 2023

Data Science Intern

Durham, NC

- Applied **HDBSCAN** clustering and **t-SNE** dimensionality reduction to **25k×25k** genomic datasets for **subgroup discovery**.
- Optimized data processing by **55%** by automating **ETL pipelines** for genomic data using **Python** and **SQL**.
- Built **interactive Tableau dashboards** to display **real-time KPIs** for stakeholders across **4 countries**.

Active Robotic Sensing Lab, NCSU

Jan 2023 - Sep 2023

Research Assistant

Raleigh, NC

- Conducted research to segment damaged and healthy leaves of the **St. Augustine turfgrass** under drought conditions.
- Implemented **HSV space** thresholding and mean clustering for segmentation and achieved an average **SSIM of 72%**.
- Utilized YOLOv8 for object detection and achieved **mAP of 89.8%**, studied data augmentation techniques and transfer learning with **ViT-16** vision transformer for semantic segmentation.

Accenture

May 2021 – June 2022

Machine Learning Engineer

Pune, India

- Improved overall performance by **15%** of an **AI-based document processing platform** serving **25+ enterprise clients**.
- Achieved **82% F1-score** in multi-class document classification by developing an **OCR & BERT-based NLP pipeline**.
- Improved email sentiment analysis model accuracy by **15%** by implementing **TF-IDF**, **WordNet**, and **Naive Bayes**.
- Developed an **NER Module** using **LayoutLMv2** to extract entities, boosting parsing accuracy by **25%**, enabling automation.
- Built an OCR-engine **recommender system** using **RandomForests**, reducing client-side processing time by **20%**.

Projects

[LLM - RL] Turning Llama 3.2-1B into a Reasoning Model (HuggingFace, vLLM, Unsloth)

Feb 2026

- Transformed Llama Instruct into a reasoning model using **GRPO** and **LoRA**, reducing trainable parameters while achieving structured **CoT** outputs on **Grade 8 Math benchmarks**.
- Engineered **4 reward functions** covering answer correctness, format compliance, and response length constraints.
- Fine-tuned and aligned a 1B-parameter LLM using vLLM-accelerated rollout generation, optimizing the full RLHF-style training pipeline with Unsloth for compute efficiency.

[NLP] LOLgorithm: Humor Classification (Python, PyTorch, TensorFlow, SciKit-Learn) [\[ArXiv\]](#)

Dec 2023

- Leveraged **CoBERT** dataset to examine the humor content in a sentence and verify the linguistic theory of humor.
- Created hand-crafted **syntactic and semantic** features modifying the embeddings from **NRClex**, **Word2Vec**, and **WordNet**
- Utilized **contextual BERT embeddings** and improved model accuracy by **14%** with all features using **Colbert model**.

[Computer Vision] Explainable AI for DeepFake Detection (Python, PyTorch) [\[GitHub\]](#)

Nov 2023

- Achieved an F1 score of **98%** with **XceptionNet** for deepfake detection on **FaceForensics++** and **Celeb-DF** datasets.
- Applied **Explainable AI (XAI)** methods such as **GradCAM**, **LIME**, and **LRP** to highlight the relevance of input to the prediction and improved transparency and interpretability.

Tehchnical Skills

Programming Languages : Python, C++, C#, R, MATLAB, SQL

Libraries : Scikit-learn, Matplotlib, Seaborn, Langchain, LlamaIndex, NLTK, SpaCy, Hugging Face Transformers, MLFlow, OpenCV, Pillow, MMCV, OpenVINO, Detectron2, AutoML, TRL, Unsloth, vLLM

Tools & Frameworks: PyTorch, TensorFlow, Keras, Databricks, PySpark, Azure, AWS, Tableau, Git, Docker, Kubernetes, HPC